

Table 4. Model Comparison: Nested Regression Models for Body Mass Prediction

Statistic

Linear Regression Models

Model 1 (Flipper Only)

Model 2 (+ Species)

Model 3 (+ Bill Measures)

Model 4 (Full Model)

R-squared

X.XXX

X.XXX

X.XXX

X.XXX

Adjusted R-squared

X.XXX

X.XXX

X.XXX

X.XXX

RMSE (g)

XXX.X

XXX.X

XXX.X

XXX.X

AIC

XXXX.X

XXXX.X

XXXX.X

XXXX.X

BIC

XXXX.X

XXXX.X

XXXX.X

XXXX.X

F-statistic

XXX.XX

XXX.XX

XXX.XX

XXX.XX

df (model, residual)

(X, XXX)

(X, XXX)

(X, XXX)

(X, XXX)

p-value (F-test)

<0.001

<0.001

<0.001

<0.001

Source: Palmer Penguins Dataset (Gorman et al., 2014)

Model 1: Body Mass ~ Flipper Length

Model 2: Body Mass ~ Flipper Length + Species

Model 3: Body Mass ~ Flipper Length + Species + Bill Length + Bill Depth

Model 4: Body Mass ~ All morphometric predictors + Species + Sex